

AUSTIN-BERGSTROM, TX, USA

WMO: 722540

Lat: 30.183N

Lon: 97.680W

Elev: 480

StdP: 14.44

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.6	29.9	12.1	10.4	39.1	16.6	12.9	42.4	25.2	47.5	23.2	49.7	6.0	0	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.4	100.3	74.3	98.4	74.6	96.5	74.7	78.7	89.2	78.1	88.5	77.5	87.6	9.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	140.7	81.0	75.8	137.4	80.6	75.1	134.3	80.4	42.5	89.5	41.8	88.7	41.2	87.5	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.0	19.4	17.6	20.4	103.7	4.1	3.3	17.5	106.1	15.1	108.0	12.8	109.9	9.8	112.3	
			DB												
			WB	18.7	81.0	3.6	2.1	16.1	82.5	14.0	83.7	12.0	84.9	9.3	86.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.8	50.9	54.9	61.2	68.5	76.2	82.2	84.4	84.9	79.5	70.1	59.9	52.1
	DBStd	14.29	9.69	10.32	8.98	7.23	5.81	3.72	3.13	3.28	5.44	7.74	9.27	9.44
	HDD50	297	108	61	17	1	0	0	0	0	0	1	22	87
	HDD65	1648	447	308	181	46	4	0	0	0	1	41	204	416
	CDD50	7154	135	197	364	557	811	966	1066	1082	884	622	319	151
	CDD65	3030	9	24	63	152	350	516	601	617	435	198	51	14
	CDH74	33510	60	172	430	1277	3318	5973	7473	7807	4669	1886	369	76
	CDH80	16400	5	35	78	371	1350	3012	4107	4409	2295	669	63	6
Wind	WSAvg	7.7	8.0	8.7	9.0	9.0	8.8	7.9	7.2	6.6	6.0	6.7	7.4	7.5
Precipitation	PrecAvg	35.4	2.8	2.0	2.6	2.8	4.2	3.0	2.6	2.0	2.9	4.6	3.0	2.7
	PrecMax	62.0	10.0	7.7	8.8	6.7	14.1	11.3	9.7	7.8	7.5	21.3	10.2	13.0
	PrecMin	17.3	0.0	0.1	0.3	0.3	0.3	0.3	0.3	0.0	0.0	0.1	0.0	0.3
	PrecStd	11.7	2.4	1.8	1.8	1.7	2.9	2.4	2.0	2.0	1.9	4.7	2.6	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.5	84.6	85.8	91.9	95.9	101.4	102.1	102.9	101.1	92.9	85.6	80.1
		MCWB	61.8	63.8	66.7	70.7	74.2	74.0	73.9	74.7	72.7	72.6	69.3	65.3
	2%	DB	75.1	79.0	82.1	87.5	92.7	97.6	99.5	100.5	96.9	89.7	81.6	76.0
		MCWB	61.5	62.9	66.1	69.6	74.1	74.8	74.4	74.6	73.9	72.2	68.3	64.8
	5%	DB	71.5	75.0	79.0	84.1	90.2	95.2	97.7	98.7	94.0	87.0	78.5	72.6
		MCWB	61.0	62.5	65.5	69.1	73.7	75.1	74.6	74.7	73.9	70.8	66.5	63.3
	10%	DB	68.1	71.3	75.5	81.3	87.4	92.8	95.7	96.6	91.3	83.9	75.0	69.2
		MCWB	60.0	62.0	64.2	67.8	73.1	75.1	74.9	74.9	73.5	69.7	65.8	62.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.6	70.0	71.5	75.4	79.0	80.0	79.3	79.6	78.9	77.7	73.8	70.4
		MCDB	72.3	75.7	79.0	84.3	87.9	91.4	91.1	90.6	88.0	84.4	79.4	74.6
	2%	WB	66.8	68.2	69.9	73.6	77.3	78.5	78.3	78.4	77.8	76.2	71.9	68.6
		MCDB	71.1	73.3	75.9	81.8	86.4	89.2	89.4	89.2	86.9	82.9	77.0	73.0
	5%	WB	64.7	66.5	68.7	72.3	76.1	77.8	77.5	77.8	77.0	75.0	70.1	66.4
		MCDB	69.5	71.4	74.4	79.9	85.0	88.2	88.1	88.6	86.0	81.6	75.2	71.0
	10%	WB	61.6	64.5	67.3	71.0	75.0	77.0	77.0	77.1	76.2	73.6	68.2	63.5
		MCDB	66.9	69.6	73.2	77.9	83.6	86.9	87.4	87.9	85.0	79.8	73.7	68.0
Mean Daily Temperature Range	5% DB	MDBR	23.0	22.1	22.2	22.0	20.2	20.3	21.6	22.4	21.9	23.0	22.7	22.6
		MCDBR	27.6	26.4	24.2	23.9	22.5	23.4	24.7	25.6	25.0	24.8	24.9	26.2
		MCWBR	16.6	15.2	12.6	10.7	7.8	5.7	5.1	5.5	7.1	10.2	13.8	16.9
	5% WB	MCDBR	20.7	19.6	19.5	18.3	18.3	18.9	20.6	21.2	19.3	17.9	18.7	20.9
		MCWBR	15.2	13.5	12.2	9.6	7.8	6.2	5.6	5.6	6.6	8.8	12.7	16.1
Clear-Sky Solar Irradiance	taub	0.334	0.341	0.363	0.405	0.450	0.432	0.445	0.433	0.414	0.378	0.365	0.322	
	taud	2.477	2.466	2.414	2.317	2.236	2.336	2.333	2.351	2.359	2.443	2.413	2.525	
	Ebn at Noon	285	293	292	282	269	272	268	271	273	277	271	284	
	Edh at Noon	29	32	36	41	45	41	40	39	38	32	31	27	
All-Sky Solar Radiation	RadAvg	953	1101	1424	1708	1832	2088	2106	1993	1644	1357	1018	842	
	RadStd	102	198	132	153	159	140	172	105	152	186	117	109	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144

Nomenclature: See separate page